

Customer Review Cleaning System

Task Overview

This project focuses on **cleaning and preparing IMDb movie reviews for NLP analysis**. Raw reviews often contain HTML tags, links, punctuation, numbers, unnecessary words, and different forms of the same word. These unwanted elements can make text analysis less effective.

The project takes the **IMDb 50K review dataset** as input and applies a series of text preprocessing techniques. The cleaned reviews are stored separately while keeping the original reviews available for comparison.

Dataset

- **Dataset:** IMDb 50K Movie Reviews
- **Columns:** review, sentiment
- **Reviews:** 50,000
- **Duplicate reviews found:** 418
- **Final reviews:** 49,582

Process

- **Data Loading:** Pandas is used to load and inspect the dataset.
- **Data Checking:** Missing values and duplicate reviews are identified and handled.
- **Text Cleaning:** Reviews are converted to lowercase and HTML tags, URLs, numbers, punctuation, and special characters are removed using **Regular Expressions**.
- **Stopword Removal:** Common English words that provide less useful information are removed using **NLTK**.
- **Tokenization:** Each review is divided into individual words.
- **Stemming:** Words are reduced to their basic word forms using **Porter Stemmer**.
- **Cleaned Column:** The processed reviews are stored in a new `cleaned_review` column.

Example

Original:

This movie was AMAZING!!!

Cleaned:

movi amaz

Output

The original and cleaned reviews are compared to verify the preprocessing. The complete processed dataset is exported as:

Result

The project converts **raw movie reviews into clean and standardized text**, making the dataset ready for further NLP applications such as **sentiment analysis, text classification, and machine learning**.